

Guiding VFX Students Through a Shifting Landscape: Balancing Foundations with Future-Facing Tools

Rico Dober

Filmuniversity Babelsberg KONRAD WOLF

Potsdam, Germany

r.dober@filmuniversitaet.de



Figure 1: Students at a Virtual Production Workshop, 2025.

Abstract

The rapid evolution of visual effects (VFX) technologies is reshaping both production pipelines and the skills expected of graduates entering the industry. Real-time rendering, virtual production, and generative artificial intelligence (AI) have expanded the creative toolkit more than ever before, but they also raise urgent pedagogical questions: How can educators prepare students for a profession where tools evolve faster than curricula? How can programs ensure graduates remain adaptable, employable, and artistically mature, rather than short-lived “tool operators”? This talk builds on the outcomes of the SIGGRAPH 2025 panels Navigating the Future of VFX Education [Dober et al. 2025] and Education Disrupted [Giri et al. 2025], as well as insights from other presentations at the conference [Gloor and Giacchino 2025] [Andreassen et al. 2025]. Both educators and industry supervisors emphasized three priorities: (1) grounding students in artistic and cinematic foundations such as composition, lighting, and storytelling; (2) ensuring proficiency in traditional VFX workflows including matchmove, compositing, and CG integration; and (3) creating structured opportunities for future-facing experimentation with real-time tools, AI-assisted workflows, and hybrid pipelines. Case studies from classroom projects demonstrate how students can begin with conventional pipelines and later pivot to emerging technologies without sacrificing production

quality or conceptual integrity. For example, projects that transitioned mid-production to real-time methods improved iteration speed while still requiring a solid grasp of cinematic fundamentals. By embedding artistic foundations, workflow literacy, and experimentation into curricula, educators can foster adaptability and critical judgment. Industry-ready graduates are not defined by the software they know today, but by their ability to evaluate, learn, and creatively apply the tools of tomorrow.

CCS Concepts

• **Applied computing** → **Media arts**; • **Computing methodologies** → **Computer graphics**; **Image manipulation**; **Artificial intelligence**; **Virtual reality**; • **Social and professional topics** → **Computing education**.

ACM Reference Format:

Rico Dober. 2025. Guiding VFX Students Through a Shifting Landscape: Balancing Foundations with Future-Facing Tools. In *SIGGRAPH Asia 2025 Educator's Forum (SA Educator's Forum '25)*, December 15–18, 2025, Hong Kong, Hong Kong. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3757372.3771854>

1 Introduction

The visual effects industry has always been highly affected by technological change, but the speed and extent of recent developments such as real-time rendering pipelines, affordable virtual production setups and AI-generated imagery prove extremely difficult for educators. Although VFX studios are actively exploring these tools, they continue to rely on traditional workflows for the majority of high-end production tasks due to legal uncertainties and quality concerns. This dual reality requires a balanced approach to teaching: one that equips students with timeless artistic and technical skills

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SA Educator's Forum '25, Hong Kong, Hong Kong

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ACM ISBN 979-8-4007-2132-8/25/12

<https://doi.org/10.1145/3757372.3771854>

while enabling them to explore new technologies in a controlled and purposeful way.

2 Challenges In The Current Educational Landscape

Students entering VFX programs today often arrive with a strong interest in the latest tools and trends, but limited awareness of the craft's foundational principles. This creates three common educational challenges:

- (1) **Premature Specialization:** Students neglect core skills in favor of niche, fast-changing tools.
- (2) **Tool Obsolescence:** By the time students graduate, some tools and techniques they learned may already be outdated.
- (3) **Industry Expectation Gap:** VFX Studios still prioritize candidates with a sound understanding of traditional workflows and problem-solving ability.

3 A Three-Pillar Framework For VFX Education

The proposed framework is designed to address these challenges by anchoring curricula around:

- **Artistic and Cinematic Foundations:** Drawing, photography, cinematography, lighting, composition, and narrative form the creative baseline for all technical skills.
- **Traditional VFX Workflows:** Understanding matchmoving, compositing, rendering passes, simulation, and asset integration ensures that students can work in current production environments.
- **Future-Facing Experimentation:** Dedicated “sandbox” courses or project modules give students the opportunity to explore Unreal Engine, Gaussian splatting, and AI-assisted concepting, while emphasizing critical evaluation of tool capabilities and limitations.

4 Case Studies

Classroom examples demonstrate how the framework works in practice:

- A short film project began in a conventional Greenscreen pipeline before transitioning mid-production to a LED Wall centered Virtual Production approach, improving quality and efficiency.
- An AI-assisted previs project accelerated concept exploration, but still relied on a solid grasp of shot composition, pacing, and narrative clarity.
- A virtual production exercise on an LED stage allowed students to integrate real-time backgrounds into a traditionally lit foreground shoot.

5 Conclusion

The role of the VFX educator is not to predict the exact tools students will use in their careers, but to ensure that graduates possess the artistic maturity, technical fluency, and adaptability to thrive in an industry that will inevitably look different in five years. By combining strong artistic foundations with workflow proficiency and structured innovation, VFX programs can produce artists who

are ready for both the demands of today and the opportunities of tomorrow.

Acknowledgments

This document was assisted by ChatGPT, a language model developed by OpenAI. The version used for this task is ChatGPT-5, as of August 2025. The tool was utilized to aid in rephrasing and summarizing text. The content generated was reviewed and edited to ensure accuracy and relevance.

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